Antony Hubert

M.Sc. Mechanical Engineer | Metal Additive Manufacturing (PBF-LB/M)

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Based in Aalen, Germany | Open to relocation worldwide
German Job Seeker Visa; eligible for EU Blue Card. Visa sponsorship welcome. Available immediately.
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PROFESSIONAL SUMMARY

Mechanical engineer specialising in metal 3D printing (laser powder bed fusion). I take difficult materials from first trials through to a documented, repeatable production process, designing the experiments, the tooling and the measurements myself. My master's research covered a magnetic alloy that industry treats as unworkable, and the results were published in a peer-reviewed journal. Before moving to Germany I worked in automotive design and CNC manufacturing in India. I am based in Aalen and open to relocating worldwide.

SELECTED ACHIEVEMENTS

Made a difficult material usable. FeSi6.5 wastes up to 30% less energy than standard electrical steel, but it is too brittle to roll or press, so industry avoids it. I developed a laser technique that made 3D-printed parts around 60% smoother while keeping them fully dense. That is precise enough for the thin layered cores used in electric motors and transformers. Published in Procedia CIRP (2024).

Turned a material property into a setting engineers can control. Repeating the laser step at chosen intervals reshaped the metal's internal grain structure on demand, with no cracking and no loss of accuracy.

Improved the machine, not only the settings. Designed, simulated, built and commissioned a gas-flow system with a heated build platform for a research printer. It went on to support the entire research programme.

CORE COMPETENCIES

Metal 3D Printing: Laser powder bed fusion (PBF-LB/M, L-PBF, SLM) and DMLS · Industrial systems: SLM 280HL, TRUMPF TruPrint 1000 · Materials: FeSi6.5, AlSi10Mg, Co-Cr · Process and parameter development · In-process laser remelting · Defect and porosity analysis · Support design · Design for Additive Manufacturing (DfAM) · Powder handling

Simulation and Testing: ANSYS Fluent (CFD) and ANSYS Mechanical (FEA) · Design of Experiments · Taguchi method · ANOVA and regression analysis

CAD and Design: CATIA V5 · Siemens NX · PTC Creo · AutoCAD · Reverse engineering · GD&T (ISO 1101) · Automotive body and EV component design · Concept sketching and CAD rendering · Training and mentoring

Laboratory Analysis: Sample preparation and metallography · Optical and scanning electron microscopy (SEM) · Atomic force microscopy (AFM) · Surface measurement (KEYENCE VR-3100, MarSurf M400) · Dilatometry and TMA (NETZSCH DIL 402, METTLER TOLEDO) · Heat treatment · Porosity and grain analysis · Micro-CT

Software: Autodesk Netfabb and Machine Control Framework · Materialise Magics · ZEISS ZEN Core · Python · Machine learning fundamentals

Manufacturing: 4-axis CNC machining and CAM · Toolpath optimisation · Process qualification · Technical documentation · Project coordination

PROFESSIONAL EXPERIENCE

Academic Research Assistant, Metal 3D Printing **09/2023 – 03/2024**

Hochschule Aalen (Aalen University), LaserApplication Center, Germany. SmartPro / Smart-ADD research programme

- Led process development for a brittle magnetic alloy from first trials to a documented, repeatable production window. Results published in a peer-reviewed journal.
- Carried out all laboratory analysis myself, from sample preparation through microscopy to surface measurement, with no external lab required. Coordinated with the team to run the experiments and deliver everything on time.
- Completed the gas-flow system begun in my working student role, adding platform heating before building, installing and commissioning it.

Working Student, Metal 3D Printing **05/2023 – 08/2023**

Hochschule Aalen (Aalen University), LaserApplication Center, Germany

- Started the gas-flow system project: gathered requirements, designed the parts in CATIA V5 and tested the design by flow simulation in ANSYS.
- Ran metal 3D printers end to end, covering design, build preparation, machine operation, finishing and inspection.
- Reduced failed builds through better support design and machine settings.

Application Engineer, CAD / CAE **03/2022 – 09/2022**

CADD Centre Training Services Pvt. Ltd., Chennai, India

- Trained and mentored over 100 engineers and students in CATIA V5, Siemens NX, PTC Creo, ANSYS and AutoCAD.
- Designed automotive body structures and electric-vehicle components, and advised clients on whether their designs

could be manufactured.

CAD / CAM Engineer, CNC Manufacturing **06/2021 – 02/2022**

Axis Automation Pvt. Ltd., Chennai, India

• Programmed and operated 4-axis CNC machines producing precision automotive parts.

• Recreated legacy parts from physical samples by reverse engineering.

• Improved cutting strategies to raise output and reduce material waste.

EDUCATION

M.Sc. Advanced Materials and Manufacturing **10/2022 – 09/2025**

Hochschule Aalen (Aalen University), Germany. Final grade 2.7 (German scale: 1.0 best, 4.0 pass)

• Focus: metal 3D printing, powder metallurgy, materials analysis and magnetic alloys.

• Thesis: improving the surface quality of 3D-printed FeSi6.5 magnetic components through in-process laser remelting. Supervisors: Prof. Dr. Harald Riegel, Prof. Dr. Dagmar Goll.

• Minor project: thermal expansion of 3D-printed AlSi10Mg with and without heat treatment, motivated by hinge clearances on satellite solar arrays. Designed a 30-specimen geometry matrix, ran the heat treatment at 300 °C, and measured expansion to 400 °C by dilatometry and TMA. Printed on an SLM 280HL. Two-person project with Anusha Arulalan; I led specimen design, heat treatment and the dilatometry campaign. Supervisor: Prof. Dr. Markus Merkel.

B.E. Mechanical Engineering **07/2017 – 08/2021**

Jeppiaar Institute of Technology, Chennai, India. CGPA 8.5 / 10 (Indian scale, 10 best)

• Thesis (lead author): reducing porosity in 3D-printed cobalt-chromium for medical implants. A structured Taguchi study identified scan speed as the deciding factor and cut porosity to 2.8%, using 9 experiments instead of 27.

PUBLICATION

Hofele M., Antony H., Kolb D., Riegel H. (2024). Production of soft magnetic FeSi6.5 parts with precise layer topography using integrated laser remelting at PBF-LB/M to induce multi-material lamination. Procedia CIRP, Volume 124, pages 6 to 11. DOI: 10.1016/j.procir.2024.08.060

CONTINUING PROFESSIONAL DEVELOPMENT

Independent Study in AI and Machine Learning for Manufacturing (2025 to present). Self-directed work in machine learning, data analysis and Python, applied to monitoring 3D-printing quality in real time. This is the next step identified in my thesis.

BEYOND ENGINEERING

• Volunteer, German Red Cross, 30 hours (2022/23), through the CONTRIBUTE programme of Aalen University's International Relations Office.

• Student Coordinator, TECH AGRONA 2K20 national project expo, Jeppiaar Institute of Technology.

• SAE reverse engineering workshop. Automotive Styling Boot Camp, 80 hours in sketching, digital sculpting, rendering and surface modelling.

CERTIFICATIONS

• Applied Ergonomics. NPTEL / IIT Madras, 2020

• CAE and CFD using ANSYS. CIPET, Chennai, 2019

• CAD/CAM using PTC Creo. CIPET, Chennai, 2019

• CAD/CAM using CATIA V5. CIPET, Chennai, 2019

LANGUAGES

• English: C1 (Advanced). Language of my Master's study, thesis and publication.

• German: B1 (Intermediate). B2 examination scheduled before end of December 2026.

• Tamil: C2 (Native / Bilingual).